

VEO Patent Strengthening Instructions

Non-Provisional Upgrade — Patent #3 — v4

Method and System for Establishing Portable, Verifiable Trust Over Non-Deterministic Computation

Date: June 28, 2026 **Scope:** Expand protection from a specific implementation into a foundational infrastructure layer. **Rule:** Do NOT rewrite the existing invention. Preserve all existing content and ADD the following concepts, embodiments, and claim coverage.

⚠ v4 Changelog (June 28, 2026)

This version incorporates strategic insights from multi-perspective patent review (USPTO examiner, Big Tech IP strategist, potential acquirer). Changes from v3:

1. **NEW: Trust Non-Amplification Invariant** — Critical anchoring concept for Patent #5
 2. **NEW: "Trust Health" vocabulary** — Establishes a new measurable property category
 3. **NEW: Verifiable Decision Object (VDO) explicitly named** — Strengthens Patent #4 continuation
 4. **NEW: Anti-101 defense guidance** — Hardware grounding for software claims
 5. **NEW: Layered claim drafting rules** — Abstract independent claims, specific dependent claims
 6. **NEW: Patent #5 continuation anchor** — Trust Runtime / Continuous Trust Health
 7. **UPDATED: Five-Layer Trust Stack** — Layer 5 now explicitly covers Trust Health monitoring
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★ The Core Reframe: Trust Over Uncertainty

This is the most important instruction in this document.

The invention is NOT entropy. The invention is **trust over uncertainty**. Those are different things.

Current Framing Throughout the Provisional: - Generate entropy - Score entropy - Verify entropy

Required Framing Throughout the Non-Provisional: - Establish trust in non-deterministic computation

The Analogy

Standard	Did NOT Invent	DID Invent
PDF	Documents	Portable trusted documents
JWT	Identity	Portable trusted identity
VEO	Randomness	Portable trusted uncertainty

Introduce: Trust Assertion

A VEO is not merely an object. It is a statement. It says:

"This output was generated using these sources, at this quality, with this signature, under this policy, at this trust level."

That statement is the invention. Introduce language such as:

A VEO constitutes a cryptographically verifiable trust assertion regarding a non-deterministic computation.

You are no longer patenting an entropy object. You are patenting a trust assertion.

Terminology Progression (Gradual, Not Abrupt)

Do NOT eliminate entropy language. Entropy is the technical foundation. Trust is the architectural purpose. Build upward.

Level	Term	Role
1	Entropy	The raw material (keep in technical description)
2	Uncertainty artifact	The structured container
3	Trust artifact	The signed, scored, verifiable container
4	Trust assertion	The statement the artifact makes about the computation

Let the specification naturally ascend: - Early sections: "generate entropy from multiple sources" - Middle sections: "construct an uncertainty artifact with provenance and scoring" - Later sections: "produce a cryptographically verifiable trust assertion" - Claims: frame around trust assertions, not entropy generation

Concrete Instructions

1. In the Abstract, augment (don't replace) entropy language with trust assertion language.
The system generates entropy AND produces trust assertions.
2. In every independent claim, build from entropy generation upward to trust establishment.
The entropy is the input. The trust assertion is the output.
3. Add a section titled "Trust Assertion" that defines a VEO as constituting a verifiable trust assertion about a non-deterministic computation.
4. Frame ECS as both entropy quality scoring AND trust quantification — broadening applicability beyond randomness.
5. Frame the verification framework as progressively evaluating the trust level of an uncertainty artifact.

The Test

A reader should finish the non-provisional believing the invention covers any system where: - An output involves uncertainty or non-determinism - A structured object captures the uncertainty's provenance and quality - Cryptographic and/or immutable proofs establish trust - The object constitutes a verifiable trust assertion - Independent verification is possible without trusting the generator

That scope includes AI decisions, autonomous systems, simulations, governance, gaming — and anything else where non-deterministic computation needs trust infrastructure.

Goal: Protect the category, not just the implementation.

1. Reframe VEO as a Universal Uncertainty Artifact

Current framing: Verifiable Entropy Object is a structured entropy object.

Expand to: A Verifiable Entropy Object (VEO) is a portable, machine-verifiable uncertainty artifact that encapsulates one or more entropy sources, provenance information, quality assessments, cryptographic attestations, trust metadata, and verification evidence.

Goal: Protect the concept of uncertainty becoming a transferable, auditable digital artifact.

2. Add AI Decision Attestation Embodiments

The provisional mentions AI in the background but does not claim it.

Add embodiments covering: LLM tool selection, agent routing, multi-agent arbitration, resource allocation, autonomous task assignment, sampling decisions, autonomous execution approval, model selection, AI governance workflows.

Example claim language:

A computer-implemented method comprising: generating a verifiable entropy object; associating the verifiable entropy object with an autonomous agent decision; recording a decision outcome linked to the verifiable entropy object; enabling independent verification of the randomness influencing the decision.

Goal: Directly cover future AI infrastructure.

3. Introduce Decision Receipt Concepts

New embodiment: Decision Receipt / Decision Attestation Record.

```
{  
  "decision_id": "...",  
  "decision_type": "tool_selection",
```

```
"selected_option": "Tool B",  
"confidence": 0.81,  
"veo_id": "...",  
"signature": "...",  
"anchor": "..."  
}
```

Covers: Human decisions, AI decisions, multi-agent decisions, governance votes, automated execution events.

Goal: Create continuation opportunities around auditable decision systems.

4. Generalize Signature Systems

Current: Ethereum, EIP-191, secp256k1.

Expand to: secp256k1, Ed25519, RSA, post-quantum signatures, future asymmetric systems.

Language: "Any asymmetric cryptographic signature scheme capable of proving authenticity and permitting verification."

Goal: Prevent easy design-around.

5. Generalize Hash Functions

Current: SHA-256.

Expand to: SHA-256, SHA-3, Keccak, Blake2, Blake3, future cryptographic hash functions.

Language: "One or more cryptographic hash functions."

Goal: Future-proof claim scope.

6. Generalize Anchoring Targets

Current: Blockchain anchoring.

Expand to: Public blockchains, consortium chains, private chains, append-only ledgers, transparency logs, notarization networks, distributed timestamping systems.

Claim: "Immutable verification substrate" instead of only "blockchain."

Goal: Protect the trust model rather than a specific ledger.

7. Verification as a Trust State Machine

Current levels: Structurally Valid, Cryptographically Verified, Anchored Verified, Policy Failed, Invalid.

Describe as: A trust state machine where an object transitions between verification states.

Additional states: pending, partially verified, expired, revoked, superseded.

Claim: "Assigning an object to one of a plurality of trust states based on progressively evaluated verification criteria."

Goal: Separate patentable value from the entropy itself.

8. ECS Beyond Fixed Weights

Add embodiments: Dynamic weighting, policy-based weighting, industry-specific weighting, machine-learned weighting, regulator-defined weighting.

Language: "The dimensions and weighting coefficients may be modified based on application requirements."

Goal: Avoid limiting protection to current percentages.

9. Interoperability Claims

Embodiments: One provider generates, another verifies, another consumes.

Key statement: "A verifier need not trust the original generator."

Goal: Standards-layer positioning.

10. Entropy-as-a-Service

Services: Entropy generation, VEO issuance, VEO verification, ECS scoring, anchoring.

Claim: "Issuing verifiable entropy objects through a network-accessible service."

Goal: Protect future API businesses.

11. Multi-Party Governance

Examples: DAO voting, lottery systems, regulated gaming, compliance approvals, corporate governance.

Pattern: VEO referenced during outcome generation; outcome remains independently auditable.

Goal: Increase licensing opportunities.

12. AI Audit Trail

Covers: Decision provenance, replayability, regulatory review, model governance, post-event investigation.

Example: "A third party may reconstruct the decision context by verifying the associated VEO and related decision records."

Goal: Position VEO as future AI governance infrastructure.

13. [NEW v4] Anti-101 Defense: Ground Software in Physical Limits

The Problem

Because these patents deal with cryptography, algorithms, and decision-making, patent examiners will reflexively try to reject them under 35 U.S.C. 101 as "abstract ideas."

The Defense

Do NOT let the examiner frame the VEO system as "organizing information" or "mathematical relationships."

The specification must emphasize that:

1. **The VDF component acts as a physical temporal barrier.** It leverages a strict sequential hardware computing constraint that cannot be broken even by infinite parallel processing. This directly improves computer network security by permanently closing a temporal system vulnerability (the preview gap).
2. **The ECS scoring system quantifies measurable physical properties** of entropy sources — freshness (time decay), diversity (source count), independence (category separation), manipulation resistance (computational difficulty). These are measurements of real-world system properties, not abstract mathematical concepts.
3. **The anchoring system creates an irreversible physical state change** on an immutable verification substrate. Once anchored, the trust assertion cannot be unwound. This is a concrete improvement to computer system reliability.
4. **The dual-path asynchronous architecture** — sub-millisecond in-memory token delivery decoupled from slow-chain Merkle proof anchoring — solves a concrete engineering trade-off between security and performance that prior art fails to address.

Claim Drafting Rule

Every independent claim should begin with a **physical or computational act** (obtaining entropy from hardware sources, computing sequential hash chains, writing to an immutable substrate) before reaching any **trust assertion** language.

Goal: Preempt 101 rejection by grounding every claim in hardware constraints and measurable physical properties.

14. [NEW v4] Layered Claim Drafting: Anti-Design-Around

The Problem

Corporate legal teams design cheap workarounds by looking at specific nouns in independent claims and replacing them with alternatives.

The Rule

Abstract functional language in independent claims. Specific implementations in dependent claims.

```
[Broad Independent Claim]
→ "An immutable verification substrate..."

├─ [Dependent Claim] → "...wherein the substrate is a public blockchain."
├─ [Dependent Claim] → "...wherein the substrate is an append-only transparency log."
└─ [Dependent Claim] → "...wherein the substrate is a distributed timestamping network."
```

Term Mapping

Independent Claim Language	Dependent Claim Specifics
"immutable verification substrate"	Polygon, Ethereum, Solana, append-only log
"cryptographic hash function"	SHA-256, SHA-3, BLAKE3
"asymmetric cryptographic signature scheme"	secp256k1, Ed25519, RSA
"sequential computational proof"	Verifiable Delay Function, iterative hash chain
"entropy source adapter"	drand beacon, block hash, QRNG service

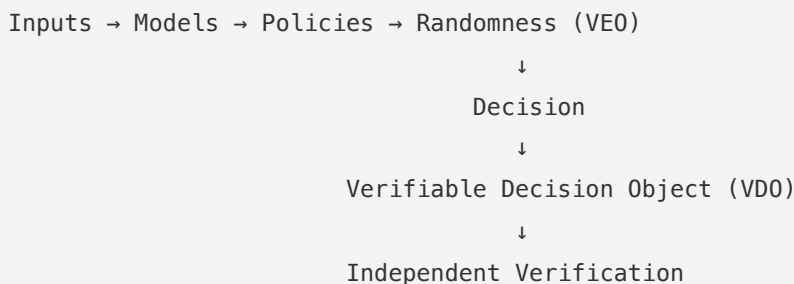
Verification

Before filing, check every independent claim. If any contains "Polygon," "SHA-256," "secp256k1," "Ethereum," "EIP-191," or any other specific implementation as a *required element*, move it to a dependent claim and replace with the generalized term.

Goal: Force competitors to infringe the broad claim regardless of which specific technology they substitute.

15. Reserve Continuation: Patent #4 — Verifiable Decision Objects (VDO)

The non-provisional should explicitly leave room for a continuation covering **Verifiable Decision Objects (VDO)**.



VEO becomes one component of a larger decision-verification stack: - VEO provides the entropy foundation - **VDO** wraps the decision that used that entropy - The VDO references the VEO, the model, the inputs, and the policy - The entire chain is independently verifiable

VDO Definition (for specification text)

A Verifiable Decision Object (VDO) is a structured, cryptographically signed digital object that encapsulates the complete context of a non-deterministic computational decision, including: the identity of the deciding entity, the input parameters, the applicable policy constraints, the referenced Verifiable Entropy Object(s), the decision outcome, and independently verifiable attestation evidence. A VDO enables post-hoc reconstruction and audit of autonomous decisions without requiring trust in the deciding entity.

[NEW v4] Trust Non-Amplification Invariant

The specification MUST include the following principle:

Trust Non-Amplification Invariant: A derivative decision object's computed trust score cannot mathematically exceed the trust score of the root entropy object(s) contributing to it. Trust is conserved or degraded through processing — never amplified. A VDO consuming a VEO with an ECS score of 720 cannot itself claim a trust score exceeding 720. This invariant ensures that the trust system is mathematically self-correcting and prevents artificial trust inflation.

This principle is critical because: 1. It distinguishes the system from standard logging (logs don't enforce trust conservation) 2. It creates a mathematically provable property that prior art cannot match 3. It anchors Patent #5's "Trust Health" monitoring on a formal foundation

Application domains for Patent #4: AI agents, autonomous vehicles, robotics, financial trading, healthcare workflows, enterprise automation.

To reserve this continuation: Ensure the non-provisional specification includes at least one embodiment describing a Verifiable Decision Object that references a VEO, introduces the Trust Non-Amplification Invariant, without fully specifying the VDO protocol. This creates the priority chain.

16. [NEW v4] Reserve Continuation: Patent #5 — Trust Runtime / Continuous Trust Health

The non-provisional should explicitly reference a system-wide trust monitoring and governance layer.

The Core Concept: Trust Health

Existing monitoring systems observe: - **Security** (Splunk, SIEM) - **Performance** (Datadog, New Relic) - **Availability** (Uptime monitors)

The Trust Runtime monitors a fundamentally different axis:

Trust Health — the continuous, quantitative measurement of computational trust quality across a system of interacting components that produce, consume, and propagate trust

assertions.

This is analogous to: - Battery Health (measures charge capacity degradation over time) - Cluster Health (measures distributed system coordination quality) - **Trust Health** (measures trust assertion quality degradation across delegation and time)

Trust Health Indicators (for specification text)

The specification should describe, in at least one embodiment, a Trust Runtime that monitors:

1. **ECS Drift** — tracking how entropy confidence scores change across a fleet of VEO-producing systems over time
2. **Source Concentration Risk** — detecting when too many trust assertions depend on too few entropy sources
3. **Chain Depth** — measuring how many delegation steps separate a trust assertion from its root entropy
4. **Anchoring Latency** — monitoring the time between VEO generation and immutable substrate anchoring
5. **Trust Score Distribution** — observing system-wide trust score patterns for anomalies
6. **Trust Non-Amplification Compliance** — verifying that no derivative objects violate the invariant

Specification Text (minimal embodiment)

In one embodiment, the system further comprises a Trust Runtime that continuously monitors trust health across a distributed network of VEO producers and consumers. The Trust Runtime tracks trust assertion quality metrics including entropy confidence score drift, source concentration risk, delegation chain depth, anchoring latency, and trust score distribution. The Trust Runtime enforces the Trust Non-Amplification Invariant across all derivative trust objects in the network and generates alerts when trust health indicators fall below configurable thresholds. The Trust Runtime enables continuous compliance verification without requiring re-audit of individual trust assertions, providing system-wide governance over computational trust quality.

Why This Matters for Filing

This text in the Patent #3 specification enables filing Patent #5 as a Continuation Application with the 2026 priority date. Without this text, Patent #5 would need to establish a new priority date, exposing it to any prior art published between now and the future filing date.

Goal: Anchor the "Trust Health" vocabulary and continuous trust monitoring concept in the earliest possible filing.

17. Named Architecture: Five-Layer Trust Stack

For long-term portfolio coherence, the specification should reference a layered architecture:

Layer	Name	Function	Patent
1	Trust Source	Generate verifiable entropy from multiple independent sources	#1
2	Trust Integrity	Prevent manipulation and preview via sequential computational proof	#2
3	Trust Evidence	Package into VEO with scoring, signing, anchoring — producing trust assertions	#3
4	Trust Coordination	Link trust artifacts to decisions via VDO; preserve trust across delegation boundaries	#4 (future)
5	Trust Runtime	Continuous Trust Health monitoring, system-wide governance, compliance enforcement	#5 (future)

Include at least one paragraph describing this layered architecture. If a company evaluates the portfolio in 5 years, they should see the beginnings of an entire infrastructure stack — not isolated patents.

[NEW v4] Architecture Properties

The specification should note the following architectural properties of the Five-Layer Trust Stack:

1. **Each layer depends on layers below it** — Trust Health monitoring (L5) is meaningless without trust assertions to monitor (L3)
2. **Trust flows upward and is conserved** — via the Trust Non-Amplification Invariant

- 3. **Verification can occur at any layer independently** — a verifier at Layer 3 need not understand Layer 5
- 4. **The stack is composable** — applications may use Layers 1-3 without Layers 4-5

Goal: Architectural coherence increases strategic value beyond the strength of any single patent.

Strategic Summary

The non-provisional should no longer read as:

"A better random number system."

It should read as:

"A system for establishing portable, verifiable trust over non-deterministic computation — across software systems, AI agents, governance systems, gaming systems, and distributed networks — with continuous trust health monitoring and governance."

Patent Family Progression

Patent	What It Protects	Category	Problem Solved
#1	Generate verifiable randomness	Randomness generation	Randomness cannot be independently verified
#2	Prevent randomness manipulation and preview	Randomness integrity	Randomness can be previewed
#3	Package into a portable, auditable trust artifact	Trust over uncertainty	Trust disappears after randomness is consumed
#4 (future)	Verify AI and automated decisions via VDO	Decision verification	Trust cannot survive delegation

Patent	What It Protects	Category	Problem Solved
#5 (future)	Continuous Trust Health monitoring and governance	Trust runtime	Trust silently decays over time

Each patent extends the previous one while opening a larger application domain. The portfolio defines five distinct properties of a computational trust system: 1. **Establishing** trust 2. **Protecting** trust 3. **Packaging** trust 4. **Preserving** trust across delegation 5. **Maintaining** trust over time

That coherence maximizes long-term patent value, continuation opportunities, licensing potential, and acquisition attractiveness.

[NEW v4] Big Tech Interest Map

For strategic context during claim drafting — understanding who values what helps prioritize claim coverage:

Buyer	Primary Interest	Why
OpenAI	#4, #5	Autonomous agents create delegation + long-running trust problems
Google Cloud	#5	Runtime governance maps to cloud ops + enterprise compliance
Microsoft	#5	Governance, compliance, enterprise tooling alignment
Gaming (GLI, BMM)	#1–#3	Easiest to commercialize today, regulatory requirement

[NEW v4] Monday Morning Directive for Saint Island

"Regarding the draft for 05_Patent3_NonProvisional.pdf:

1. Ensure all independent claims utilize generalized functional terminology — 'immutable verification substrate' not 'blockchain', 'cryptographic hash function' not 'SHA-256', 'asymmetric cryptographic signature scheme' not 'secp256k1'.
2. Ground the 101 defense on hardware computational bottlenecks of sequential hashing and physical temporal barriers.
3. Explicitly include the Five-Layer Trust Stack architecture description with all five layers named.
4. Add the Trust Non-Amplification Invariant definition.
5. Add the Verifiable Decision Object (VDO) definition and at least one Decision Attestation embodiment with JSON structure.
6. Add the Trust Health / Trust Runtime embodiment paragraph for Patent #5 continuation anchoring.
7. Verify that no specific implementation names (Polygon, Ethereum, SHA-256, secp256k1, EIP-191) appear in any independent claim — move all to dependent claims.

Please confirm once these anchors have been added to the working draft."

Reference Documents

#	Document	Purpose
1	Patent3_VEO_Provisional.pdf	Filed provisional application
2	RFC-0001-VEO1.md	Frozen protocol specification
3	ECS-v1.md	Frozen scoring specification
4	OpenRNG_Whitepaper_v1.md	Category-defining whitepaper
5	Agent Arbiter demo	Working AI decision attestation example

#	Document	Purpose
6	Patent3_NonProvisional_Instructions_v3.md	Previous version of these instructions
7	Patent portfolio strategic review (June 28, 2026)	Multi-perspective analysis informing v4 changes